

Karunakara Reddy Julakanti

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[LinkedIn](#)

Full Stack Engineer with 3+ years of experience in scalable backend development using Node.js and frontend development with React.js. Proficient in AWS cloud, microservices, and RESTful APIs. Experienced in integrating LLMs, LangChain, RAG, and vector databases (FAISS) into real-world applications. Hands-on with PyTorch, TensorFlow, and end-to-end model fine-tuning. Strong in serverless architecture, performance optimization, and AI-driven solutions.

Work Experience:

Data Engineer, Horizon IT, , NC

July 2025 - Present

LLM Fine-Tuning, Vector Search & Cloud Migration:

- Fine-tuned LLMs using custom datasets for domain-specific QA, increasing model accuracy by 30%.
- Applied FAISS for high-speed vector search in RAG pipelines, reducing query response times by 50%.
- Built and deployed deep learning models using PyTorch and TensorFlow, achieving >90% accuracy in classification tasks.
- Migrated 5+ TB of data from Teradata to BigQuery, leveraging GCP services (Dataflow, Cloud Storage, Composer) to modernize legacy architecture and reduce operational costs by 40%.

Software Engineer, Concordia University Wisconsin, Mequon, WI

January 2024 - May 2025

AI Dashboards & Microservice Architecture:

- Developed interactive dashboards using React.js, Plotly.js improving user reporting speed by 35%.
- Built scalable REST APIs with Node.js and MongoDB, handling over 10K+ data transactions daily.
- Integrated LangChain + RAG with LLMs to create intelligent assistants, boosting internal document query resolution rate by 65%.
- Automated deployments with Docker on AWS EC2, reducing release times by 40%.

Software Engineer, Couth Infotech, Hyderabad, India

April 2022 - December 2024

Serverless Web Application Development:

- Delivered a full-stack serverless app with React.js, AWS Lambda, and DynamoDB, reducing infrastructure cost by 30%.
- Designed modular APIs with Express.js and InversifyJS, improving code reusability by 50%.
- Achieved 99.9% uptime by implementing fault-tolerant AWS infrastructure with minimal ops overhead.

Education:

Master of Science, Computer Engineering, GPA: 3.5, Concordia University Wisconsin, Mequon, WI .

- Awarded University Graduate Scholarship.

Projects:

RAG-based Q&A System

- Built document-based QA system using **LangChain**, **Qdrant**, **FastEmbed**, **Flashrank**, **ChatGroq**
- Parsed and embedded Meta's Q1 2024 report for semantic search and intelligent answering

AI Customer Support Agent

- Created RAG pipeline with **FastEmbed**, **SQLite**, and **Groq AI** for 24/7 automated query resolution

Fake News Detector

- Fine-tuned **BERT** for binary classification using real/fake news dataset to identify misinformation

Technical Skills:

JavaScript, Typescript, SQL, Python, C, C++, AWS(S3, Lambdas, EC2, EKS, DynamoDB), Deep Neural Networks(RNN, CNN), LLMs, GCP (BigQuery, Composer, GCS, Dataflow, Dataproc)